2.1 Major steps in ETL process Data Extraction Methods

Major steps in ETL process

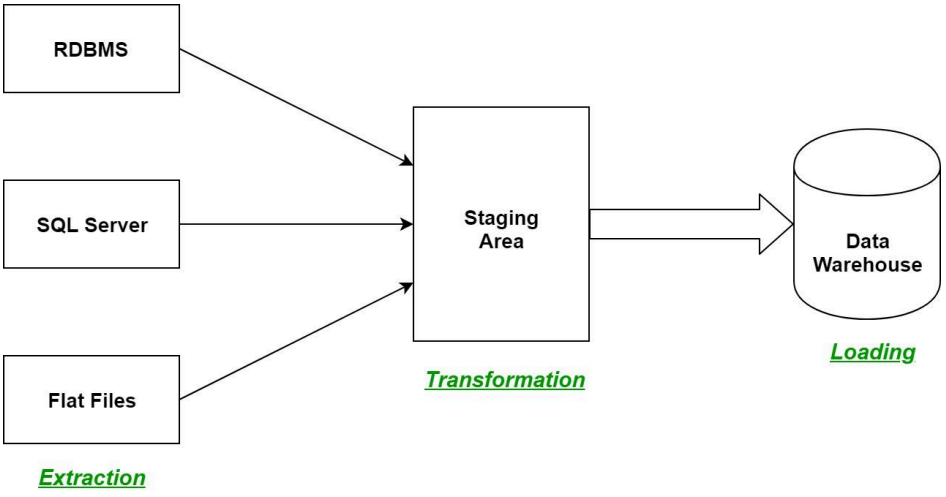
Answer

INTRODUCTION:

1. ETL stands for Extract, Transform, Load and it is a process used in data warehousing to extract data from various sources, transform it into a format suitable for loading into a data warehouse, and then load it into the warehouse. The process of ETL can be broken down into the following three stages:
2. **Extract**: The first stage in the ETL process is to extract data from various sources such as transactional systems, spreadsheets, and flat files. This step involves reading data from the source systems and storing it in a staging area.
3. **Transform**: In this stage, the extracted data is transformed into a format that is suitable for loading into the data warehouse. This may involve cleaning and validating the data, converting data types, combining data from multiple sources, and creating new data fields.
4. **Load**: After the data is transformed, it is loaded into the data warehouse. This step involves creating the physical data structures and loading the data into the warehouse.
5. The ETL process is an iterative process that is repeated as new data is added to the warehouse. The process is important because it ensures that the data in the data warehouse is accurate, complete, and up-to-date. It also helps to ensure that the data is in the format required for data mining and reporting.

Additionally, there are many different ETL tools and technologies available, such as Informatica, Talend, DataStage, and others, that can automate and simplify the ETL process.

ETL is a process in Data Warehousing and it stands for **Extract**, **Transform** and **Load**. It is a process in which an ETL tool extracts the data from various data source systems, transforms it in the staging area, and then finally, loads it into the Data Warehouse system.



Let us understand each step of the ETL process in-depth:

1. **Extraction:**

   The first step of the ETL process is extraction. In this step, data from various source systems is extracted which can be in various formats like relational databases, No SQL, XML, and flat files into the staging area. It is important to extract the data from various source systems and store it into the staging area first and not directly into the data warehouse because the extracted data is in various formats and can be corrupted also. Hence loading it directly into the data warehouse may damage it and rollback will be much more difficult. Therefore, this is one of the most important steps of the ETL process.

2. **Transformation:**

   The second step of the ETL process is transformation. In this step, a set of rules or functions are applied on the extracted data to convert it into a single standard format. It may involve following processes/tasks:

- Filtering – loading only certain attributes into the data warehouse.
- Cleaning – filling up the NULL values with some default values, mapping U.S.A, United States, and America into USA, etc.
- Joining – joining multiple attributes into one.
- Splitting – splitting a single attribute into multiple attributes.
- Sorting – sorting tuples on the basis of some attribute (generally key-attribute).

3. **Loading:**

   The third and final step of the ETL process is loading. In this step, the transformed data is finally loaded into the data warehouse. Sometimes the data is updated by loading into the data warehouse very frequently and sometimes it is done after longer but regular intervals. The rate and period of loading solely depends on the requirements and varies from system to system.

Advantages of ETL process in data warehousing:

1. **Improved data quality:** ETL process ensures that the data in the data warehouse is accurate, complete, and up-to-date.
2. **Better data integration:** ETL process helps to integrate data from multiple sources and systems, making it more accessible and usable.
3. **Increased data security:** ETL process can help to improve data security by controlling access to the data warehouse and ensuring that only authorized users can access the data.
4. **Improved scalability: E**TL process can help to improve scalability by providing a way to manage and analyze large amounts of data.
5. **Increased automation:** ETL tools and technologies can automate and simplify the ETL process, reducing the time and effort required to load and update data in the warehouse.

Disadvantages of ETL process in data warehousing:

1. **High cost:** ETL process can be expensive to implement and maintain, especially for organizations with limited resources.
2. C**omplexity:** ETL process can be complex and difficult to implement, especially for organizations that lack the necessary expertise or resources.
3. **Limited flexibility:** ETL process can be limited in terms of flexibility, as it may not be able to handle unstructured data or real-time data streams.

4. **Limited scalability**: ETL process can be limited in terms of scalability, as it may not be able to handle very large amounts of data.
5. **Data privacy concerns**: ETL process can raise concerns about data privacy, as large amounts of data are collected, stored, and analyzed.

éData Extraction Methods

https://www.tutorialspoint.com/what-are-the-various-techniques-for-data-extraction
https://www.tutorialspoint.com/what-are-the-various-extraction-methods-in-data-warehouses

Data extraction is the process of analyzing and crawling through data sources (such as databases) to recover vital information in a specific pattern. Data is processed further, including metadata and other data integration; this is another step in the data workflow.

Unstructured data sources and various data formats account for most data extraction. Tables, indexes, and analytics can all be used to store unstructured data.

Data in a warehouse can come from various places, and a data warehouse must use three different approaches to use it. Extraction, Transformation, and Loading are the terms for these procedures (ETL).

Data extraction entails retrieving information from disorganized data sources. The data extracts are subsequently imported into the relational Database's staging area. The source system is queried for data via application programming interfaces, and extraction logic is applied. Due to this process, the data is now ready to go through the transformation phase of the ETL process.

The extraction method in data warehousing depends on the source rules and business requirements. The amount of data to be extracted and the ETL phase (initial load or ongoing maintenance) also influence the extraction method. There are two main types of extraction methods: Logical Extraction Methods and Physical Extraction Methods.

Logical Extraction Methods:
1. Full Extraction: All data is extracted entirely from the source system, with no need to track changes.
2. Incremental Extraction: Only data changed since a specific event (e.g., last extraction) is extracted, requiring a way to identify these changes, often using timestamps or change logs.

Physical Extraction Methods:
1. Online Extraction: Data is extracted directly from the source system, either from source tables or intermediate systems (e.g., snapshot logs).
2. Offline Extraction: Data is extracted from an external source, such as redo logs or tablespaces, often created by an extraction routine.

2.2 Data Transformation; Basic Tasks in Transformation, Major Data Transformation Types

Data Transformation;Basic Tasks in Transformation

The second step of the ETL process is transformation. In this step, a set of rules or functions are applied on the extracted data to convert it into a single standard format. It may involve following processes/tasks:

- Filtering – loading only certain attributes into the data warehouse.
- Cleaning – filling up the NULL values with some default values, mapping U.S.A, United States, and America into USA, etc.
- Joining – joining multiple attributes into one.
- Splitting – splitting a single attribute into multiple attributes.
- Sorting – sorting tuples on the basis of some attribute (generally key-attribute).

éMajor Data Transformation Types

In ETL (Extract, Transform, Load), data transformation is a crucial step where raw data extracted from various sources undergoes changes to meet the requirements of the target system or database. Here are some major data transformation types commonly used in ETL processes:

1. Filtering: Filtering involves selecting specific rows or columns from the source data based on defined criteria. This helps in removing unnecessary data and focusing only on relevant information.

2. Sorting: Sorting rearranges the rows of data in a specified order, such as ascending or descending, based on one or more columns. This ensures data consistency and facilitates easier analysis.

3. Joining: Joining combines data from multiple sources based on a common field or key. Common types of joins include inner join, outer join (left, right, full), and cross join. Joining is essential for consolidating data spread across different sources.

4. Aggregation: Aggregation involves summarizing data by performing mathematical operations such as sum, average, count, minimum, or maximum on a group of rows. It helps in reducing data volume and generating meaningful insights from large datasets.

5. Normalization/Denormalization: Normalization organizes data into multiple related tables to reduce redundancy and dependency, whereas denormalization combines normalized data back into fewer tables for improved query performance.

6. Data Cleansing: Data cleansing (or data scrubbing) involves identifying and correcting errors or inconsistencies in the source data, such as missing values, duplicate records, or incorrect formats. Clean data ensures accuracy and reliability in downstream processes.

7. Data Deduplication: Data deduplication removes duplicate records from the dataset, ensuring data integrity and eliminating redundancy. This is especially important when merging data from multiple sources.

8. Data Enrichment: Data enrichment involves enhancing the source data with additional information from external sources, such as appending demographic data, geolocation data, or market trends. It enriches the dataset for better analysis and decision-making.

9. Data Transformation Functions: These include applying various functions or expressions to manipulate data, such as string manipulation functions, mathematical functions, date/time functions, and conditional logic functions.

10. Data Format Conversion: Data format conversion involves converting data from one format to another, such as from CSV to JSON, XML to relational format, or unstructured text to structured format, to meet the requirements of the target system.

These data transformation types can be combined and customized based on specific business needs and data integration requirements in ETL processes.

2.3 Data Loading Techniques

Loading is the ultimate step in the ETL process. In this step, the extracted data and the transformed data are loaded into the target database. To make the data load efficient, it is necessary to index the database and disable the constraints before loading the data. All three steps in the ETL process can be run parallel. Data extraction takes time and therefore the second phase of the transformation process is executed simultaneously. This prepared the data for the third stage of loading. As soon as some data is ready, it is loaded without waiting for the previous steps to be completed.

The **loading** process is the physical movement of the data from the computer systems storing the source database(s) to that which will store the data warehouse database. The entire process of transferring data to a data warehouse repository is referred to in the following ways:

1. **Initial Load**: For the very first time loading all the data warehouse tables.
2. **Incremental Load**: Periodically applying ongoing changes as per the requirement. After the data is loaded into the data warehouse database, verify the referential integrity between the dimensions and the fact tables to ensure that all records belong to the appropriate records in the other tables. The DBA must verify that each record in the fact table is related to one record in each dimension table that will be used in combination with that fact table.
3. **Full Refresh**: Deleting the contents of a table and reloading it with fresh data.

Refresh versus Update

After the initial load, the data warehouse needs to be maintained and updated and this can be done by the following two methods:

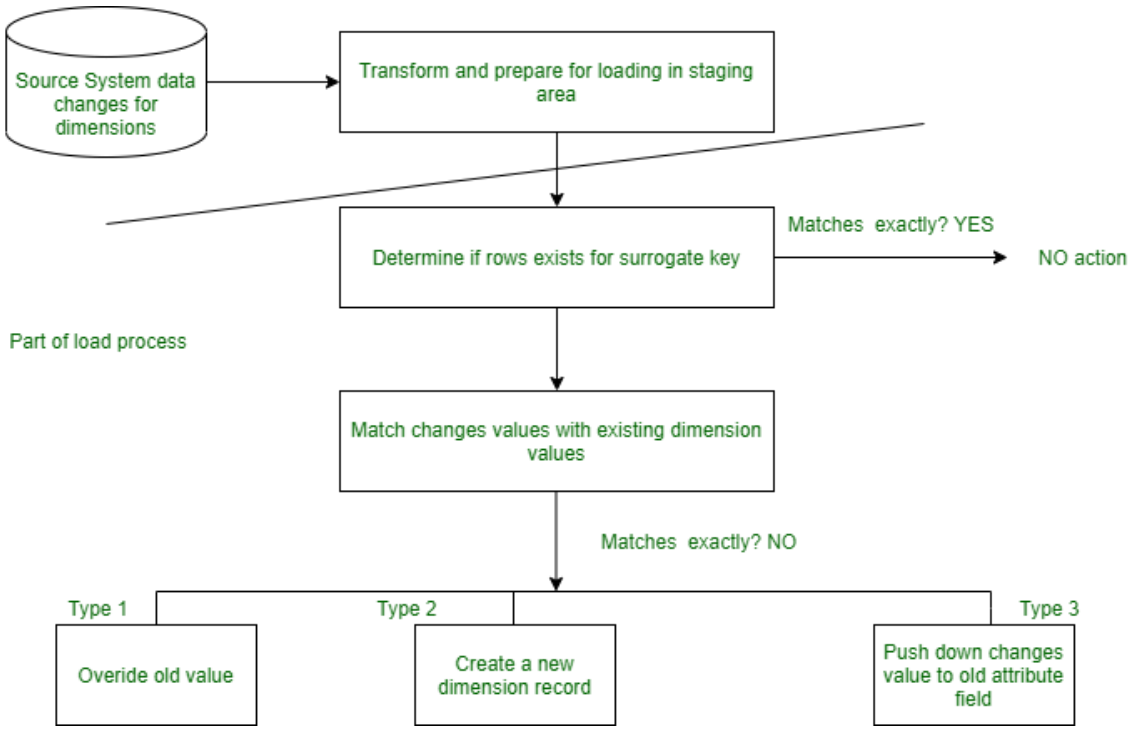
- **Update**– application of incremental changes in the data sources.
- **Refresh**– complete reloads at specified intervals.

Data Loading-

Data is physically moved to the data warehouse. The loading takes place within a “load window. The tendency is close to real-time updates for data warehouses as warehouses are growing used for operational applications.

Loading the Dimension Tables

Procedure for maintaining the dimension tables includes two functions, initial loading of the tables and thereafter applying the changes on an ongoing basis System geared keys are used in a data warehouse. The reeds in the source system have their own keys. Therefore, before an initial load or an ongoing load, the production keys must be co to system-generated keys in the data warehouse, Another issue is related to the application of ***Type 1***, **Type 2**, and ***Type 3*** changes to the data warehouse. *Fig*. shows how to handle it.



Loading changes to a dimension table

Loading the Fact tables: History and Incremental Loads

- The key in the fact table is the concatenation of keys from the dimension tables.
- So for this reason amplitude records are loaded first.
- A concatenated key is created from the keys of the corresponding dimension tables.

Methods for data loading

- **Cloud-based:** ETL solutions in the cloud are frequently able to process data in real-time and are designed for speed and scalability. They also contain the

vendor's experience and ready-made infrastructure, which may offer advice on best practices for each organization's particular configuration and requirements.

- **Batch processing:** Data is moved every day or every week via ETL systems that use batch processing. Large data sets and organizations that don't necessarily require real-time access to their data are the greatest candidates for it.
- **Open-source:** Since their codebases are shared, editable, and publicly available, many open-source ETL systems are extremely affordable. Despite being a decent substitute for commercial solutions, many tools may still need some hand-coding or customization.

Data loading challenges

Numerous ETL solutions are cloud-based, which is responsible for their speed and scalability. But large enterprises with traditional, on-premises infrastructure and data management processes often use custom-built scripts to collect and load their data into storage systems through customized configurations.

Slow down analysis: Every time a data source is added or changed, the system has to be reconfigured, which is time-consuming and hinders the ability to make quick decisions..

Increase the likelihood of errors: Changes and reconfigurations open the door to human error, duplicate or missing data, and other problems.

Require specialized knowledge: In-house IT teams often lack the necessary skills (and bandwidth) to code and monitor ETL tasks.

Require costly equipment: In addition to investing in the right human resources, organizations have to procure, house and maintain the hardware and other equipment to drive the process on site.

2.4 What is Multidimensional Data, OLAP Models: MOLAP, ROLAP.

https://www.javatpoint.com/data-warehouse-what-is-multi-dimensional-data-model
https://www.geeksforgeeks.org/multidimensional-data-model/

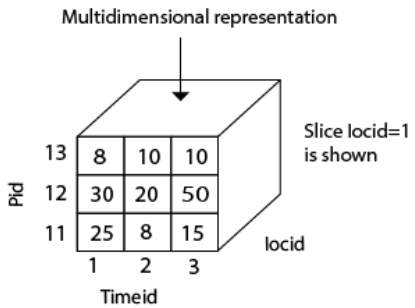
A multidimensional model views data in the form of a data-cube. A data cube enables data to be modeled and viewed in multiple dimensions. It is defined by dimensions and facts.

The dimensions are the perspectives or entities concerning which an organization keeps records. For example, a shop may create a sales data warehouse to keep records of the store's sales for the dimension time, item, and location. These dimensions allow the save to keep track of things, for example, monthly sales of items and the locations at which the items were sold. Each dimension has a table related to it, called a dimensional table, which describes the dimension further. For example, a dimensional table for an item may contain the attributes item_name, brand, and type.

A multidimensional data model is organized around a central theme, for example, sales. This theme is represented by a fact table. Facts are numerical measures. The fact table contains the names of the facts or measures of the related dimensional tables.

Tabular representation

Pid	Timeid	locid	Sales
11	1	1	25
11	2	1	8
11	3	1	15
12	1	1	30
12	2	1	20
12	3	1	50
13	1	1	8
13	2	1	10
13	3	1	10
11	1	2	35



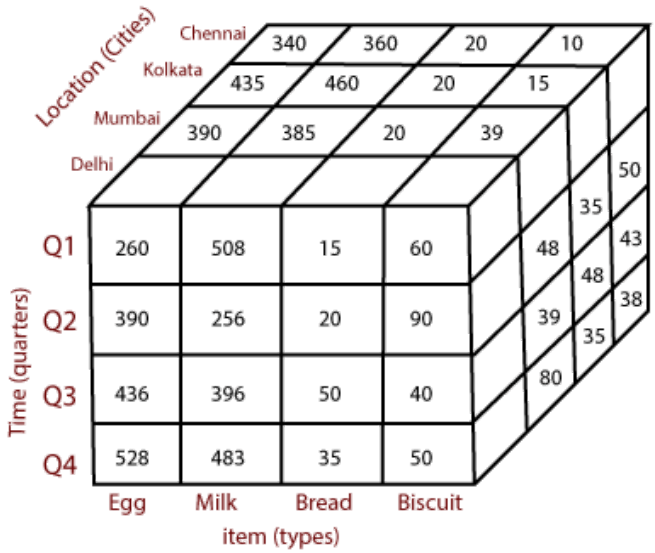
Consider the data of a shop for items sold per quarter in the city of Delhi. The data is shown in the table. In this 2D representation, the sales for Delhi are shown for the time dimension (organised in quarters) and the item dimension (classified according to the types of an item sold). The fact or measure displayed in rupee_sold (in thousands).

Location="Delhi"				
	item (type)			
Time (quarter)	Egg	Milk	Bread	Biscuit
Q1	260	508	15	60
Q2	390	256	20	90
Q3	436	396	50	40
Q4	528	483	35	50

Now, if we want to view the sales data with a third dimension, For example, suppose the data according to time and item, as well as the location is considered for the cities Chennai, Kolkata, Mumbai, and Delhi. These 3D data are shown in the table. The 3D data of the table are represented as a series of 2D tables.

	Location="Chennai"				Location="Kolkata"				Location="Mumbai"				Location="Delhi"			
	item				item				item				item			
Time	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit
Q1	340	360	20	10	435	460	20	15	390	385	20	39	260	508	15	60
Q2	490	490	16	50	389	385	45	35	463	366	25	48	390	256	20	90
Q3	680	583	46	43	684	490	39	48	568	594	36	39	436	396	50	40
Q4	535	694	39	38	335	365	83	35	338	484	48	80	528	483	35	50

Conceptually, it may also be represented by the same data in the form of a 3D data cube, as shown in fig:



OLAP is considered **(Online Analytical Processing)** which is a type of software that helps in analyzing information from multiple [databases](#) at a particular time. OLAP is simply a [multidimensional data model](#) and also applies querying to it.

Types of OLAP Servers

- Relational OLAP
- Multi-Dimensional OLAP
- Hybrid OLAP
- Transparent OLAP

https://www.geeksforgeeks.org/types-of-olap-systems-in-dbms/

MOLAP (Multidimensional Online Analytical Processing):

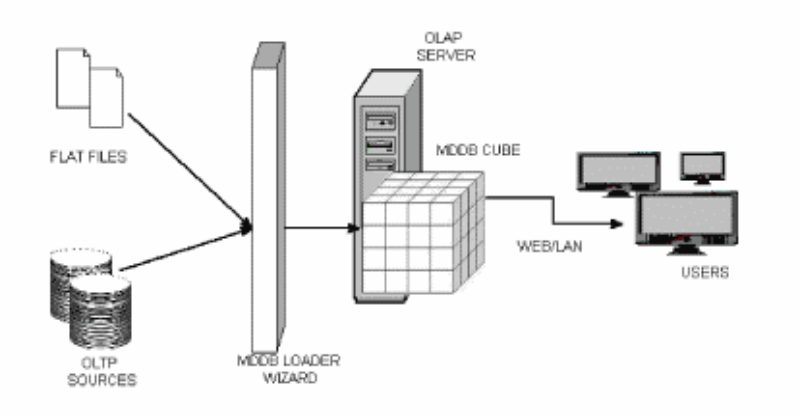
MOLAP is a type of OLAP (Online Analytical Processing) that uses multidimensional data cubes for organising and storing data. It typically involves pre-aggregating and storing data in a format optimised for fast query performance. MOLAP databases store data in a multidimensional array, which facilitates quick retrieval and analysis.

Advantages of MOLAP:

1. Optimal for slice and dice operations
2. Performs better than ROLAP when data is dense
3. Can perform complex calculations
4. Availability of extensive libraries of complex functions for OLAP analyses.

Disadvantages of MOLAP:

1. Limited Scalability
2. Can handle limited amount of data
3. Difficult to change dimension without re-aggregation.
4. Lacks security and administration features which RDBMSs can bring.



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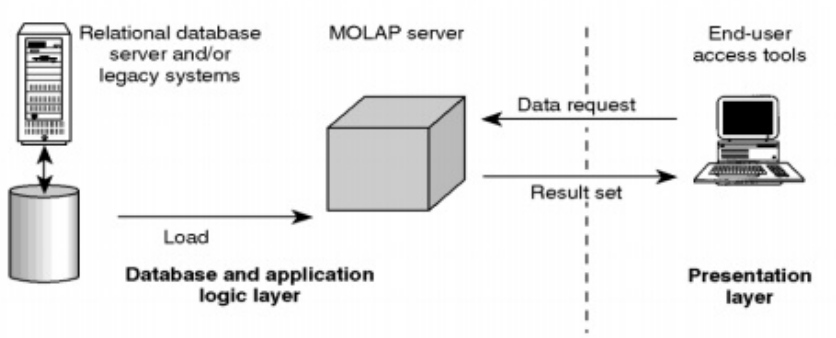

Figure Describes The Relation of the MOLAP with the server and end user.

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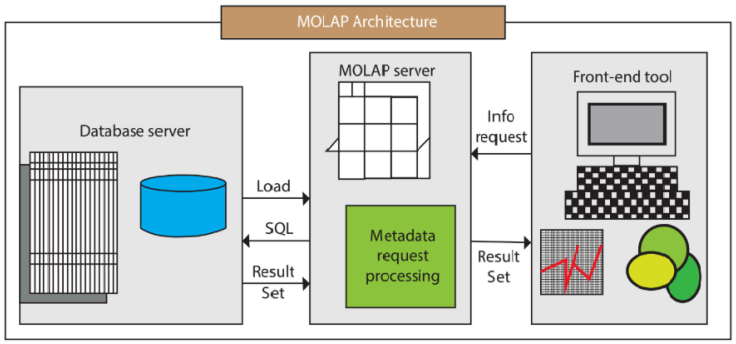


ROLAP (Relational Online Analytical Processing):

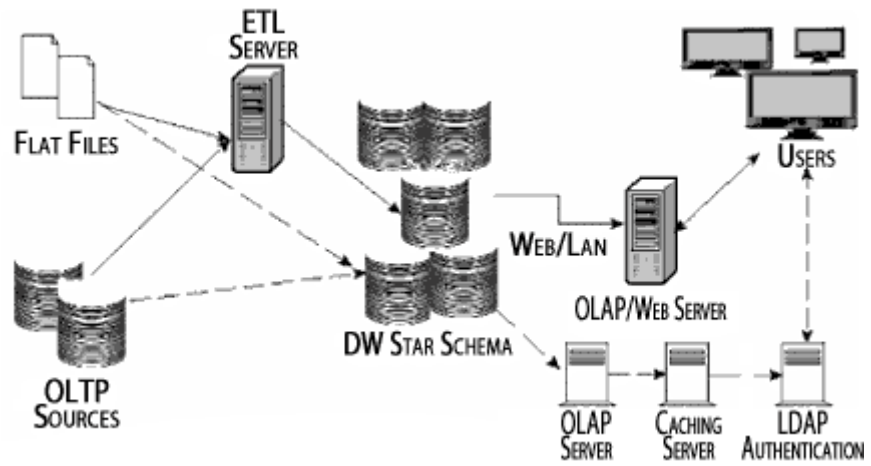
ROLAP is another type of OLAP that relies on relational databases to store and manage data. Unlike MOLAP, ROLAP does not pre-aggregate data into multidimensional cubes. Instead, it performs calculations on the fly, utilising the underlying relational database structure.

Advantages of ROLAP:

1. Can handle large amounts of data
2. Performs better than MOLAP when the data is sparse.
3. Can be used with data warehouse and OLTP systems.
4. Full security and administration is provided through RDBMSs.

Disadvantages of ROLAP:

1. Limited by SQL functionality.
2. Hard to maintain aggregate tables in the data warehouse.
3. Does not have complex functions that are provided by OLAP tools.



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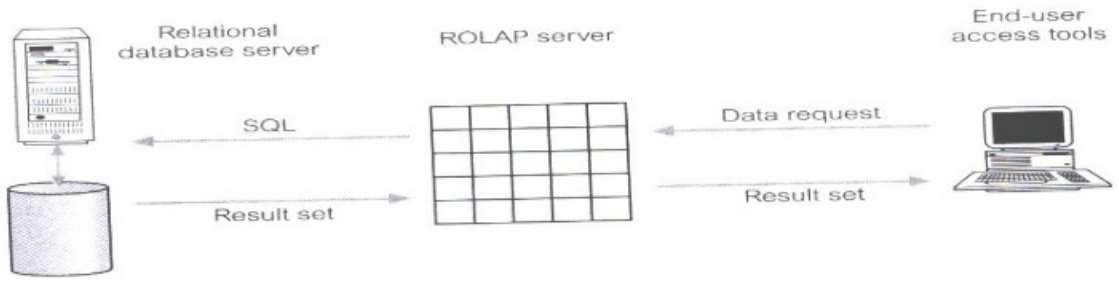

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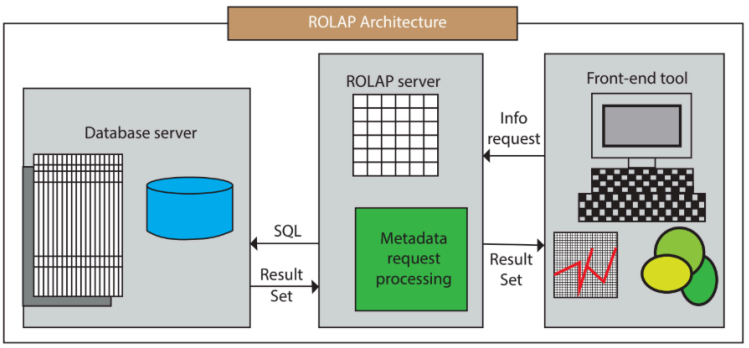


Extra

ROLAP (Relational Online Analytical Processing):

- **Data Storage:** Relies on relational databases and uses their structure for data storage.
- **Data Aggregation:** Does not pre-aggregate data; calculations are performed on-the-fly during queries.
- **Scalability:** Generally more scalable than MOLAP, capable of handling large volumes of data.
- **Flexibility:** Offers flexibility in data modeling by utilizing the existing relational database schema.
- **Query Performance:** May have slower query performance, especially for complex analytical queries.
- **Real-Time Data:** Provides potential for real-time access to data since calculations are performed on the fly.

MOLAP (Multidimensional Online Analytical Processing):

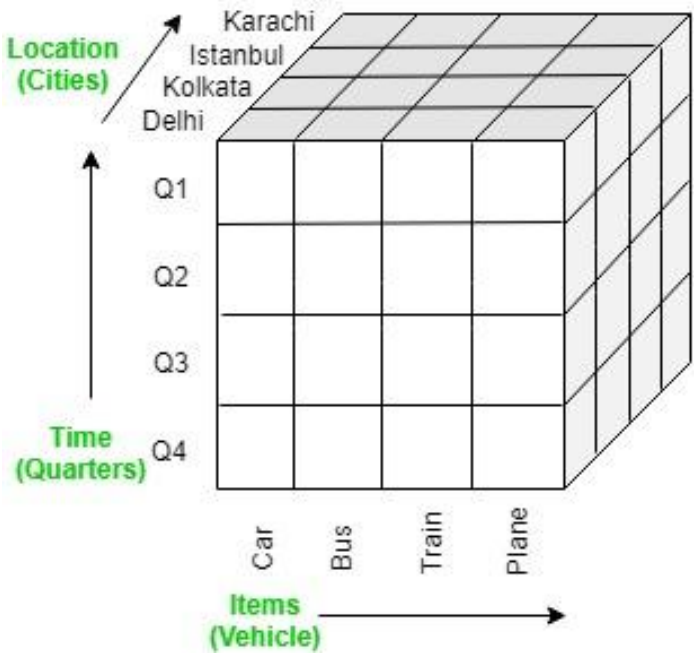
- **Data Storage:** Stores data in multidimensional cubes, pre-aggregating it for optimized query performance.
- **Data Aggregation:** Pre-aggregates data at various levels, stored in a cube structure for faster queries.
- **Scalability:** May face challenges with scalability, especially as data volumes increase.
- **Flexibility:** Structure is less flexible than ROLAP, as it is based on predefined multidimensional cubes.
- **Query Performance:** Generally offers faster query performance, well-suited for interactive and ad-hoc analysis.
- **Data Loading:** Data loading can be time-consuming, involving pre-aggregation and cube building.

2.5 OLAP operations: Slice, Dice, Rollup, Drilldown and Pivot

1)OLAP OPERATIONS

answer

OLAP stands for ***Online Analytical Processing*** Server. It is a software technology that allows users to analyse information from multiple database systems at the same time. It is based on multidimensional data model and allows the user to query on multi-dimensional data (eg. Delhi -> 2018 -> Sales data). OLAP databases are divided into one or more cubes and these cubes are known as *Hyper-cubes*.

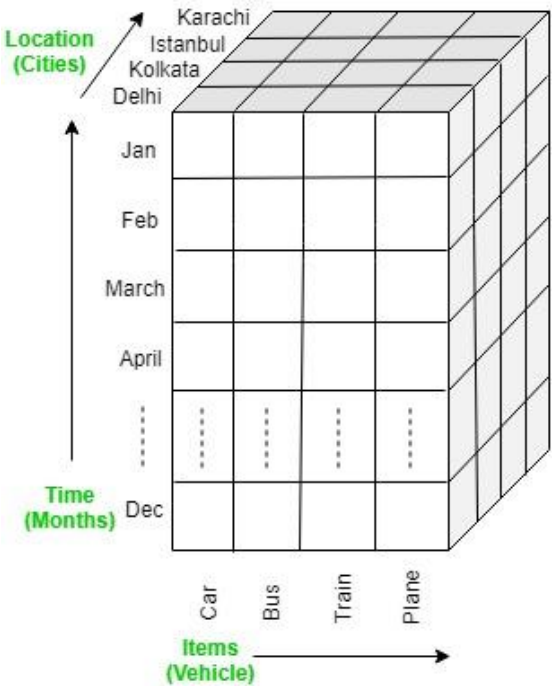


OLAP operations:

There are five basic analytical operations that can be performed on an OLAP cube:

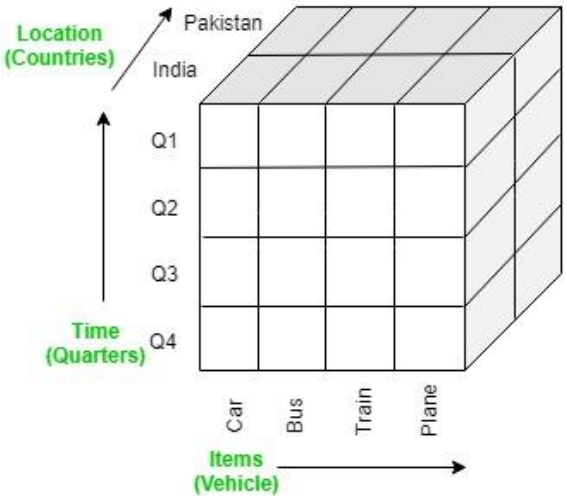
1. **Drill down:** In drill-down operation, the less detailed data is converted into highly detailed data. It can be done by:
    - Moving down in the concept hierarchy
    - Adding a new dimension

In the cube given in overview section, the drill down operation is performed by moving down in the concept hierarchy of *Time* dimension (Quarter -> Month).

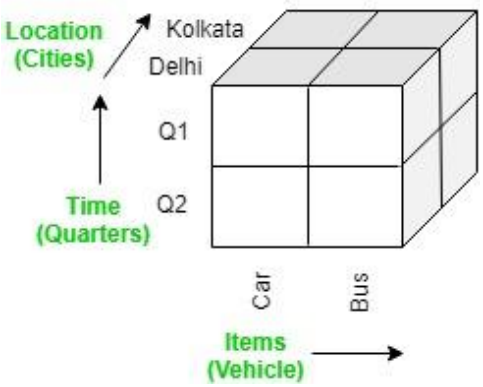


2. **Roll up:** It is just opposite of the drill-down operation. It performs aggregation on the OLAP cube. It can be done by:
    - Climbing up in the concept hierarchy
    - Reducing the dimensions

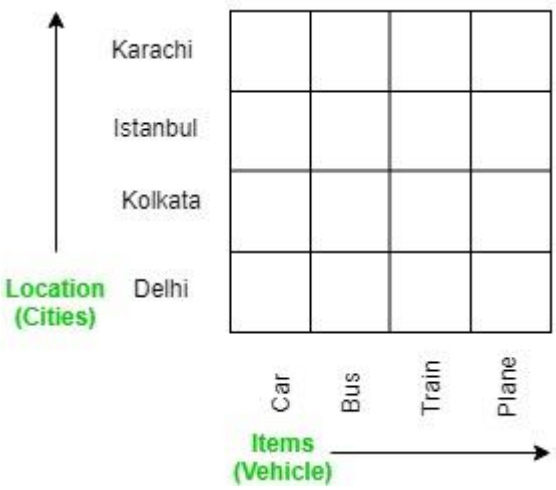
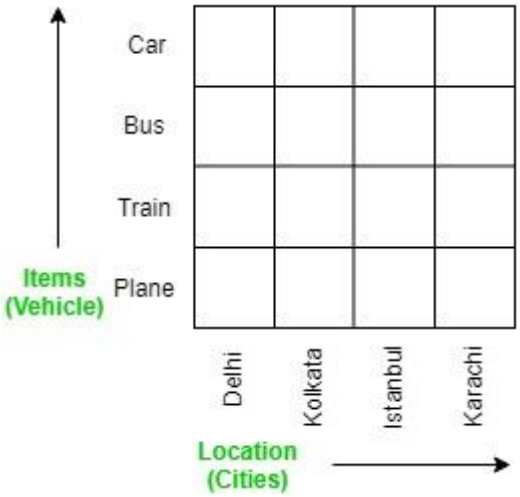
In the cube given in the overview section, the roll-up operation is performed by climbing up in the concept hierarchy of *Location* dimension (City -> Country).



3. **Dice:** It selects a sub-cube from the OLAP cube by selecting two or more dimensions. In the cube given in the overview section, a sub-cube is selected by selecting following dimensions with criteria:
    - Location = “Delhi” or “Kolkata”
    - Time = “Q1” or “Q2”
    - Item = “Car” or “Bus”



4. **Slice:** It selects a single dimension from the OLAP cube which results in a new sub-cube creation. In the cube given in the overview section, Slice is performed on the dimension Time = “Q1”.



5. **Pivot:** It is also known as *rotation* operation as it rotates the current view to get a new view of the representation. In the sub-cube obtained after the slice operation, performing pivot operation gives a new view of it.